

Trigonometry point-free

Steven Vickers

School of Computer Science, University of Birmingham

`s.j.vickers@cs.bham.ac.uk`

December 15, 2025

Abstract

Working in point-free topology under the ultraminimalist constraints of geometric logic, we develop the trigonometric functions $\sin$ and $\cos$, and prove the usual algebraic and differentiation rules.

Maths Subject Classification MSC2020 26E40; 26A24, 26A36, 18F10

Keywords: point-free topology, geometric logic

1 Introduction

This paper is part of a programme of developing real analysis in point-free topology, using the geometric style (in the sense of geometric logic) outlined in [Vic22]: all constructions are to be geometric, and, consequently, all maps are continuous. As far as possible, the aim is to use arguments already familiar from classical point-set topology, and the most pervasive differences are the need to make a careful distinction between 1-sided reals (lower or upper) and 2-sided (Dedekind), and an unexpectedly prominent role for hyperspaces (powerlocales).

Here we develop the trigonometric functions $\sin$ and $\cos$. We take the view that their fundamental characterization is as group homomorphisms $\langle \cos, \sin \rangle$ from $(\mathbb{R}, +)$ to the circle group S^1 as unit circle in $\mathbb{C} = \mathbb{R}^2$. Which homomorphism we use depends on the choice of unit for angular measurements, but already this characterization captures the angle sum rules

$$\begin{aligned}\cos(\theta + \phi) &= \cos \theta \cos \phi - \sin \theta \sin \phi \\ \sin(\theta + \phi) &= \sin \theta \cos \phi + \cos \theta \sin \phi,\end{aligned}\tag{1}$$

as well as some simpler identities: $\sin 0 = 0$, $\cos 0 = 1$, $\sin(-\theta) = -\sin(\theta)$ and $\cos(-\theta) = \cos(\theta)$.

We shall use the technology of midpoint algebras (see [Vic17] for a point-free account) to get arcs of the circle as iterative midpoint algebras (Theorem 20) and thus define (Definition 29) portions of a homomorphism defined on compact intervals. We then use the homomorphism property to extend to the whole of $\mathbb{R}$ (Theorem 30).

We start with the revolution (2π radians) as our angular unit. To define π and move to radians we use integration (Definition 25) to get $\arcsin$ as a partial inverse of $\sin$, and then the Fundamental Theorem of Calculus [Vic23] gives us the derivatives of $\sin$ and $\cos$ (Theorem 34).

The arguments presented here may seem surprisingly complicated. Some of the issues are in fact already present in a classical treatment, and are described quite carefully by G.H. Hardy in his *Pure Mathematics* [Har38]. In §163 he discusses “an unproved assumption ... that *angles are capable of numerical measurement*.” On reflection this is a little surprising. Every child is familiar with the idea of wrapping a string round a cylinder, and schoolchildren are generally happy with protractors. Why should such a familiar idea be mathematically difficult?

Hardy prefers to measure an angle as the area of its sector: this is because the arc length depends on a theory of line integral that is not developed in his book. It is not developed here either, but nonetheless we shall find a way (using the theory of midpoint algebras) to measure angles by arc length.

1.1 Foundational issues

The arguments presented here are intended to be valid in an *ultramini*malist foundations in which constructions are restricted to those that are preserved by pullback along geometric morphisms (continuous maps). It goes beyond minimalist foundations of constructive mathematics such as that of [Sam25], in that it does not even admit all the intuitionistic connectives.

The ultramini

malism comes in two flavours. The stronger, *geometric*, flavour (stronger logic, weaker constraints to satisfy) is that of geometric mathematics, relative to a given elementary topos $\mathcal{S}$ with $\mathbf{nno}$. In other words it is that fragment of the internal mathematics of bounded $\mathcal{S}$ -toposes that is preserved by inverse image functors.

The weaker, *arithmetic*, flavour is that of the internal mathematics of arithmetic universes.

I believe that the arguments here all fulfil the requirements of geometricity. Moreover, the central tool for ensuring geometricity independent of base $\mathcal{S}$ is to reason with arithmetic constructions: finite limits, *finite* colimits, and free algebra constructions. (“Arbitrary” colimits depend on $\mathcal{S}$.) Thus arithmeticity is largely baked in.

However, there is one place (see Remark 17) where I am not convinced the arithmeticity is fully justified. Our understanding of arithmetic universes is still advancing, and I hope that future results, analogous to those of [MV12], will show that the arguments presented here are arithmetically applicable.

2 Background on midpoint algebras

We recall the definitions from [Vic17], themselves adapted from [ES01].

Definition 1. A *midpoint algebra* is a space A equipped with a morphism $m: A \times A \rightarrow A$ satisfying the following conditions:

$$\begin{aligned} \text{idempotency:} \quad & m(x, x) = x \\ \text{commutativity:} \quad & m(x, y) = m(y, x) \\ \text{transposition:} \quad & m(m(x, y), m(z, w)) = m(m(x, z), m(y, w)) \end{aligned}$$

A homomorphism of midpoint algebras is a map that preserves the midpoint operation.

A midpoint algebra is *cancellative* if it satisfies

$$m(x, z) = m(y, z) \implies x = y.$$

Note that the third axiom (in the case $x = y$) tells us that $m(x, -)$ is a midpoint homomorphism.

Definition 2. A midpoint algebra A is *iterative* if, for every space X and pair of maps $h: X \rightarrow A$, $t: X \rightarrow X$ (*head* and *tail*), there is a unique morphism $M: X \rightarrow A$ making $M(x) = m(h(x), M(t(x)))$ – in other words, the following diagram commutes.

$$\begin{array}{ccc} A \times X & \xrightarrow{A \times M} & A \times A \\ \langle h, t \rangle \uparrow & & \downarrow m \\ X & \xrightarrow{M} & A \end{array}$$

Definition 3. An *interval object* I is a free iterative midpoint algebra over 2. That is to say:

1. I is equipped with two points x_- and x_+ (its *endpoints*).
2. I is an iterative midpoint algebra.
3. For every iterative midpoint algebra A with points a_- and a_+ there is a unique midpoint homomorphism $N: I \rightarrow A$ that takes x_- and x_+ to a_- and a_+ respectively.

The main result of [Vic17] is that the point-free closed Euclidean interval $\mathbb{I} = [-1, 1]$, with $m(x, y) = \frac{x+y}{2}$ and endpoints -1 and 1 , is a cancellative interval object.¹

Lemma 4. Let $f: \mathbb{I} \rightarrow \mathbb{I}$ preserve the endpoints, and be such that if $f(\theta) = f(\phi)$ then $\theta = \phi$. Then f is a homeomorphism.

¹Note that our definition of “interval object” is slightly different from that of [ES01]. On the one hand, we don’t assume that is cancellative; but on the other we expect it to be initial amongst *all* the doubly pointed iterative midpoint algebras, not just the cancellative ones. Since our $\mathbb{I}$ is cancellative, the result in [Vic17] is slightly stronger than that conjectured in [ES01].

Proof. First, we show that f is strictly monotone. For a binary relation R on $\mathbb{I}$, we write $\mathbb{I}_R$ for the corresponding subspace of $\mathbb{I}^2$.

If $\theta < \phi$ then immediately we have $f(\theta) \neq f(\phi)$, so $f \times f$ maps $\mathbb{I}_<$ to $\mathbb{I}_\neq$. We have that $\mathbb{I}_<$ is connected, $\mathbb{I}_\neq$ has two connected components $\mathbb{I}_<$ and $\mathbb{I}_>$, and $f(-1) < f(1)$, so we can deduce that $f(\theta) < f(\phi)$ whenever $\theta < \phi$. (See [Vic09a] for more on the point-free treatment of connectedness.)

We now define an inverse g for f . For $z:\mathbb{I}$, define

$$\underline{g}(z) = \{q:\mathbb{Q} \mid q < -1 \vee (q < 1 \wedge f(i(q)) < z)\},$$

where $i: \{q \in \mathbb{Q} \mid -1 \leq q < 1\} \rightarrow \mathbb{I}$ is the obvious map. This is a lower real in $\overrightarrow{[-1, 1]}$. To see why it is rounded, observe that the condition $f(i(q)) < z$ is a geometric property of $i(q)$, ie an open containing it, and so contains an open interval round it, which contains a rational bigger than q .

Similarly we define upper reals $\bar{g}(z) = \{q:\mathbb{Q} \mid (-1 < q \wedge z < f(i(q))) \vee 1 < q\}$. We show that $(\underline{g}(z), \bar{g}(z))$ is a Dedekind section $g(z)$. Disjointness is clear. For locatedness, suppose $r < s$ are rationals. Then $f(r) < f(s)$, so either $f(r) < z$ or $z < f(s)$. From the definition it easily follows that f and g are mutually inverse. $\square$

3 Arcs as midpoint algebras

We shall be working in the circle group S^1 , the unit circle in $\mathbb{C} = \mathbb{R}^2$ under the usual complex multiplication. We shall adopt standard $x + iy$ notation for complex numbers, with $\Re$ and $\Im$ for the real and imaginary parts, and shall also write $\text{sq}: S^1 \rightarrow S^1$ for the squaring map.

There we shall want to define midpoints so that (anticipating the $re^{i\theta}$ notation) the midpoint of $e^{i\theta}$ and $e^{i\phi}$ is $e^{i(\theta+\phi)/2}$. Of course, we cannot do that on the whole circle, as θ and ϕ are only defined modulo 2π , and so we get two candidate midpoints with no continuous way to choose between them. We must restrict to arcs.

Definition 5. Suppose we have $\alpha:S^1$ such that $\Im\alpha \geq 0$. Then we define the *arc* Arc_α to be the space of $z:S^1$ such that $\Re\alpha < \Re z$, and the *closed arc* $\overline{\text{Arc}}_\alpha$ to be the similar space but with $\Re\alpha \leq \Re z$.

They are the arcs from α to its complex conjugate α^* via 1. If $\alpha:\overline{\text{Arc}}_i$ and $z:\text{Arc}_\alpha$, then $-\Im\alpha < \Im z < \Im\alpha$. Also, Arc_{-1} is the open complement of the closed point -1 .

Lemma 6. Let $\alpha_\lambda:\overline{\text{Arc}}_i$ with $\Im\alpha_\lambda \geq 0$ ($\lambda = 1, 2$), and suppose $z_\lambda:\text{Arc}_{\alpha_\lambda}$. Then $z_1 z_2:\text{Arc}_{\alpha_1 \alpha_2}$.

Proof. Write $z_\lambda = x_\lambda + iy_\lambda$. Then $x_1 x_2 > \Re\alpha_1 \Re\alpha_2$ and $y_1 y_2 < \Im\alpha_1 \Im\alpha_2$, so $\Re(z_1 z_2) = x_1 x_2 - y_1 y_2 > \Re(\alpha_1 \alpha_2)$. $\square$

Corollary 7. If $\alpha:\overline{\text{Arc}}_i$ with $\Im\alpha \geq 0$, then $\text{sq}: \text{Arc}_\alpha \rightarrow \text{Arc}_{\alpha^2}$.

Definition 8. We define the *square root map* $\sqrt{\cdot}: \text{Arc}_{-1} \rightarrow \text{Arc}_i$ by

$$\sqrt{(x + iy)} = \frac{\sqrt{1+x}}{\sqrt{2}} + i \frac{y}{\sqrt{2}\sqrt{1+x}}.$$

Lemma 9. *If $z: \text{Arc}_{-1}$, then $\sqrt{z}$ is the unique point $w: S^1$ such that $w^2 = z$ and $\Re w \geq 0$.*

Proof. It's routine to show that $\sqrt{z}$ does have all those properties – in fact, $\Re w > 0$.

For uniqueness, given $z = x + iy$, suppose $w = u + iv$ has the properties. From $w^2 = z$ we have $u^2 - v^2 = x$ and $2uv = y$. The property we are trying to prove (that u and v have the values given in the statement) is a subspace, so it suffices to prove in the complementary cases $u = 0$ and $u \neq 0$.

We can rule out $u = 0$, for that would give $y = 0$, so $x = 1$ ($x = -1$ is impossible) and $v^2 = -1$.

From the equations we deduce $4u^4 - (2uv)^2 = 4u^2x$, so $(2u^2 - x)^2 = x^2 + y^2 = 1$ and $2u^2 - x = \pm 1$. If $2u^2 - x = -1$ then $x = 1$ and $u = 0$ – contradiction. Hence $2u^2 - x = 1$ and we quickly get that $w = \sqrt{z}$. $\square$

Lemma 10. *If $\alpha: \text{Arc}_{-1}$ with $\Im \alpha \geq 0$, and $z: \text{Arc}_\alpha$, then $\sqrt{z}: \text{Arc}_{\sqrt{\alpha}}$.*

Proof. Immediate. $\square$

Lemma 11. *If $z_1, z_2: \text{Arc}_i$, then $\sqrt{z_1}\sqrt{z_2} = \sqrt{(z_1 z_2)}$.*

Proof. By Lemmas 6 and 10, we know that $z_1 z_2: \text{Arc}_{-1}$, $\sqrt{z_i}: \text{Arc}_{\sqrt{i}}$, and $\sqrt{z_1}\sqrt{z_2}: \text{Arc}_i$. By Lemma 9 it remains to check $(\sqrt{z_1}\sqrt{z_2})^2 = z_1 z_2$, which is immediate. $\square$

Definition 12. The *midpoint map* $m: \text{Arc}_{-1}^2 \rightarrow \text{Arc}_{-1}$ is defined by $m(z_1, z_2) = \sqrt{z_1}\sqrt{z_2}$ (the geometric mean).

Proposition 13.

1. *The map m makes Arc_{-1} a cancellative midpoint algebra.*
2. *Each Arc_α or (for $\alpha \neq -1$) $\overline{\text{Arc}_\alpha}$ is a midpoint subalgebra.*
3. *The maps $\sqrt{\cdot}$ and sq make Arc_α and $\text{Arc}_{\sqrt{\alpha}}$, and similarly $\overline{\text{Arc}_\alpha}$ and $\overline{\text{Arc}_{\sqrt{\alpha}}}$, isomorphic as midpoint algebras.*

Proof. (1) Referring to Definition 1, the first axiom $m(z, z) = \sqrt{z}\sqrt{z} = z$ follows from Lemma 9, and the second is immediate because S^1 is Abelian.

For the third, we use Lemmas 10 and 11 to see

$$\begin{aligned} \sqrt{(\sqrt{z_1} \cdot \sqrt{z_2})} \cdot \sqrt{(\sqrt{z_3} \cdot \sqrt{z_4})} &= \sqrt{\sqrt{z_1} \cdot \sqrt{\sqrt{z_2} \cdot \sqrt{\sqrt{z_3} \cdot \sqrt{\sqrt{z_4}}}}} \\ &= \sqrt{(\sqrt{z_1} \cdot \sqrt{z_3})} \cdot \sqrt{(\sqrt{z_2} \cdot \sqrt{z_4})}. \end{aligned}$$

For cancellativity, if $m(x, y) = m(x, z)$, ie $\sqrt{x}\sqrt{y} = \sqrt{x}\sqrt{z}$, then $\sqrt{y} = \sqrt{z}$ and $y = z$.

(2) follows from Lemma 10.

(3) is clear, given the preceding lemmas. $\square$

4 Closed arcs as iterative midpoint algebras

Our plan is to exploit the fact that $\mathbb{I}$ is an interval object in order to get midpoint homomorphisms from $\mathbb{I}$ to Arc_{-1} , but for this we need their codomains to be iterative.

We shall (Theorem 20) show iterativity for arcs $\overline{\text{Arc}}_\alpha$ and Arc_α . Of course we expect that for closed arcs, because they are homeomorphic to $\mathbb{I}$, but we have not proved that yet. In order for our plan to work, we must find an independent way to prove iterativity.

The proof in [Vic17] relies on the fact that $\mathbb{I}$ is a complete metric space and that the midpoint operation halves distances.

It is clear that we do have a homeomorphism $\mathfrak{I}: \overline{\text{Arc}}_i \rightarrow [-1, 1]$, with inverse $y \mapsto \sqrt{1-y^2} + iy$, although it is not a midpoint homomorphism. It also restricts to smaller arcs $\overline{\text{Arc}}_\alpha \cong [-\mathfrak{I}\alpha, \mathfrak{I}\alpha]$, and for α close to 1 it is approximately a midpoint homomorphism. Our aim is to adapt the metric space argument for $\mathbb{I}$, using α close to 1 to control how midpoint interacts with the metric.

First, note that, as described above, $\overline{\text{Arc}}_\alpha \cong [-\mathfrak{I}\alpha, \mathfrak{I}\alpha]$. We shall write $\mathbb{I}_\alpha$ for this interval. The proof of [Vic17, Proposition 4.1] shows that $\mathbb{I}_\alpha$ is the localic completion of D_α , the set of dyadic rationals in the range $(-\mathfrak{I}\alpha, \mathfrak{I}\alpha)$ with the usual metric.

Our argument is broadly similar to that of [Vic17], but we shall use the upper hyperspace (powerlocale) $P_U \overline{\text{Arc}}_\alpha$ rather than a ball domain. This ends up simpler, because more theory is available for P_U than for the ball domain.

Briefly, the points of $P_U X$ are the compact fitted subspaces K of X . A subbase of opens is given by $\square U$, comprising those K that are inside the open U . For further details, see, eg, [Vic97].

Definition 14. Let X_0 be a metric space (or even a generalized metric space in the sense of [Vic05, Vic09b]), and let $X = \overline{X_0}$ be its localic completion. We define a *radius* map r from $P_U X$ to the space $[0, \infty]$ of upper reals by

$$rK = \inf\{q: \mathbb{Q} \mid \exists x \in X_0. K: \square B_q(x)\}.$$

(Recall that the formal balls $B_q(x)$, with $x \in X_0$ and $0 < q \in \mathbb{Q}$, form a base of opens for $\overline{X_0}$.)

Definition 15. Let A be a midpoint algebra. We define $m': A \times P_U A \rightarrow P_U A$ by $m'(a, -) = P_U(m(a, -))$.²

Lemma 16. Let A_0 be a metric space whose localic completion $A = \overline{A_0}$ is a compact midpoint algebra, and suppose we have $\kappa: [0, 1)$ such that $r(m'(w, K)) \leq \kappa r(K)$ for all $w: A, K: P_U A$.

²Fixing a means that we are working in sheaves over A , and internally there $m(a, -)$ is a map from A to A (externally, a bundle map from $A \times A$ to itself, mapping (a, b) to $(a, m(a, b))$). The geometricity of P_U (see [Vic04]) allows us to apply the internal functor P_U to this internal map, to get an internal map $P_U(m(a, -)): P_U A \rightarrow P_U A$. Externally the bundle map provides us with a map $A \times P_U A \rightarrow P_U A$.

A purely external construction would be to compose the strength $A \times P_U A \rightarrow P_U A$ with $P_U(m)$.

Then A is iterative.

Proof. [Vic17] covers the case of $[-1, 1]$, for which we have $r(m'(a, K)) = \frac{1}{2}r(K)$. We can relax the argument there.

Suppose we have maps $h: X \rightarrow A$ and $t: X \rightarrow X$. We require a unique morphism $M: X \rightarrow A$ making the diagram of Definition 2 commute.

We approximate M using maps to the upper hyperspace $P_U A$. $\mathbf{Loc}(X, P_U A)$ is a dcpo, and by compactness of A it has a bottom point $\perp$, corresponding to A as a compact subspace of itself (see [Vic95]). We define a Scott continuous endofunction T on $\mathbf{Loc}(X, P_U A)$ by $T(f) = m' \circ (A \times f) \circ \langle h, t \rangle$:

$$\begin{array}{ccc} A \times X & \xrightarrow{A \times f} & A \times P_U A \\ \langle h, t \rangle \uparrow & & \downarrow m' \\ X & \xrightarrow{T(f)} & P_U A \end{array}$$

Let M be its least fixpoint, $\bigcup_n^\uparrow M_n$ where M_0 is constant $\perp$ and $M_{n+1} = T(M_n)$. Then $r \circ M \leq \kappa(r \circ M)$, from which it follows that $r \circ M = 0$ and M factors via A thus giving us existence of the required M .

For uniqueness, suppose M' is another such. Then $M \sqsubseteq M'$ since M is least fixpoint, but the specialization order on A is discrete. $\square$

Remark 17. There is a foundational issue here – see Section 1.1. It is not clear (to me) how one can justify the existence of the least fixpoint $\bigcup_n^\uparrow M_n$ arithmetically.

There are two steps. First we construct a diagram of points (of $P_U A$) over the poset $(\mathbb{N}, \leq)$, and then use that to construct a map from the ideal completion $\text{Idl}(\mathbb{N})$ to $P_U A$. The join required is the image of the maximal ideal $\mathbb{N}$.

The second step is a consequence of Diaconescu's Theorem, that the topos of copresheaves classifies the flat presheaves. We do not yet have an arithmetic universe version of that, but it seems plausible that it can be done.

What is less clear is how one constructs the $\mathbb{N}$ -indexed diagram of points. On the face of it, relative to X , we have an endomap of $P_U A$, and so a chain of points defined inductively. The problem is that the universal characterization of $\mathbb{N}$ is with respect not to *spaces* but to *sets* (discrete spaces). With toposes we can use the discrete coreflection (“set of points”), but that is not available arithmetically.

Lemma 18. *Let z_1, z_2 be points of Arc_i , and let $z_0 = m(z_1, z_2) = \sqrt{z_1} \sqrt{z_2}$. Let $\beta = z_1 z_0^{-1}$, so multiplication by z_0 translates $\{\beta, 1, \beta^*\}$ to $\{z_1, z_0, z_2\}$. If $\Re \alpha > 0$ with $\Im \alpha \geq 0$, and $z_1, z_2: \text{Arc}_\alpha$, then*

$$2\Re \alpha |\Im \beta| \leq |\Im z_1 - \Im z_2| \leq 2|\Im \beta|.$$

Proof. We calculate

$$\Im z_1 - \Im z_2 = \Im(z_0 \beta) - \Im(z_0 \beta^*) = 2\Im \beta \Re z_0,$$

and the result follows from the fact that $\Re \alpha < \Re z_0 \leq 1$. $\square$

Lemma 19. *Let A be the midpoint algebra $\overline{\text{Arc}}_\alpha$, a localic completion via its homeomorphism with $\mathbb{I}_\alpha$, and suppose $\Re\alpha > 0$. If $K:P_U A$, and $w:A$, then $r(m'(w, K)) \leq \kappa r(K)$ where $\kappa = 1/(\sqrt{2}\sqrt{1 + \Re\alpha}\Re\alpha)$.*

Proof. Suppose $K:\Box B_q(y)$. We want $m'(w, K):\Box B_{q'}(y')$ for some $q' < \kappa q$ and some y' .

Since $\Re\alpha > 0$, we can suppose $y + q < 1$ and $-1 < y - q$. Let z_1, z_2 be the points of Arc_i corresponding to $y + q$ and $y - q$, and let z_0 and β be as in Lemma 18.

Now let $z'_\lambda = m(w, z_\lambda)$ and let us seek to apply Lemma 18 to z'_1 and z'_2 . We already have $z'_0 = m(z'_1, z'_2)$; let us take also $\beta' = z'_1(z'_0)^{-1}$. Then

$$\beta' = (\sqrt{w}\sqrt{z_1})(\sqrt{w}\sqrt{z_0})^{-1} = \sqrt{z_1}(\sqrt{z_0})^{-1}$$

and it follows that $\beta' = \sqrt{\beta}$. From Definition 8 we see that

$$|\Im\beta'| = |\Im\beta|/(\sqrt{2}\sqrt{1 + \Re\beta}) \leq |\Im\beta|/(\sqrt{2}\sqrt{1 + \Re\alpha}).$$

Now let $y'_\lambda = \Im z'_\lambda$ ($\lambda = 1, 2$). Using Lemma 18 twice, for (z_1, z_2) and for (z'_1, z'_2) , we have

$$|y'_1 - y'_2| \leq 2|\Im\beta'| \leq 2|\Im\beta|/(\sqrt{2}\sqrt{1 + \Re\beta}) \leq \kappa|\Im z_1 - \Im z_2| = 2\kappa q.$$

□

Theorem 20. *Suppose $\Im\alpha \geq 0$.*

1. *If $\alpha \neq -1$ then $\overline{\text{Arc}}_\alpha$ is an iterative midpoint subalgebra of Arc_{-1} .*
2. *For any α , Arc_α is an iterative midpoint subalgebra of Arc_{-1} .*

Proof. (1) By Proposition 13, $\overline{\text{Arc}}_\alpha$ is isomorphic as midpoint algebra to $\overline{\text{Arc}}_{\sqrt{\alpha}}$, so it suffices to prove that $\overline{\text{Arc}}_{\sqrt{\alpha}}$ is iterative. Hence, we can assume without loss of generality that $\Re\alpha > 1/\sqrt{2}$, and $\kappa < 1$ where κ is as in Lemma 19. We can now apply Lemma 16.

(2) Arc_α is isomorphic as midpoint algebra to $\text{Arc}_{\sqrt{\alpha}}$, so it suffices to assume $\alpha \neq -1$.

If we have $\langle h, t \rangle: X \rightarrow \text{Arc}_\alpha \times X$, then, by embedding Arc_α in $\overline{\text{Arc}}_\alpha$, we get $M: X \rightarrow \overline{\text{Arc}}_\alpha$ as in Definition 2. If $x:X$, then $M(x) = m(h(x), M(t(x)))$, and this is in Arc_α because $h(x)$ is. Hence M factors via Arc_α . □

The argument for part (1) of the theorem uses compactness of $\overline{\text{Arc}}_\alpha$, and one might wonder whether there is a more general argument that also covers Arc_α directly. As evidence that compactness of $\overline{\text{Arc}}_\alpha$ might be important for iterativity of Arc_α , note that $\mathbb{R}$ is not iterative. Take $X = \mathbb{N}$, and $\langle h, \text{suc} \rangle: \mathbb{N} \rightarrow \mathbb{R} \times \mathbb{N}$ where $h(n) = 2^n$. If M exists then $M(n+1) = 2M(n)$. This is because if we replace h by $h_1(n) = 2^{n+1}$, then clearly M would be doubled; but we can get the same effect by replacing $\mathbb{N}$ by the natural numbers starting at 1. Hence $M(0) = (1 + 2M(0))/2$, contradiction. (Alternatively, look at the formula $M(n) = \sum_{j=n}^{\infty} h(j)/2^{j-n+1}$ in [Vic17]. This clearly diverges for our choice of h .) If one compactifies $\mathbb{R}$, with $\pm\infty$, one no longer has a midpoint algebra.

Definition 21. Using Theorem 20, let

$$e_{\frac{1}{8}}: \left[-\frac{1}{8}, \frac{1}{8}\right] \rightarrow \overline{\text{Arc}}_{\sqrt{i}}$$

be the unique midpoint homomorphism mapping $\pm\frac{1}{8}$ to $\sqrt{(\pm i)}$.

It is calculating cos and sin for angles measured in revolutions (so in the range $\pm 45^\circ$).

5 Quasihomomorphisms

Here we bring together some miscellaneous results concerning situations where a map would be a group homomorphism, but for the fact that its domain is not closed under the group multiplication.

Definition 22. Let $0 < b: \mathbb{R}$, and $f: [-b, b] \rightarrow S^1$. Then f is a *quasihomomorphism* if

- $f(0) = 1$, and
- if $\theta, \phi, \theta + \phi$ are all in $[-b, b]$, then $f(\theta + \phi) = f(\theta)f(\phi)$.

Clearly, the property of being quasihomomorphism is preserved by composition with endomorphisms of S^1 (such as raising to a power).

Lemma 23. Let $f_1: [-b, b] \rightarrow S^1$ be a quasihomomorphism. Then f_1 extends uniquely to a quasihomomorphism $f_2: [-2b, 2b] \rightarrow S^1$.

Proof. If the extension exists, then $f_2(\theta) = f_2(\theta/2)^2 = f_1(\theta/2)^2$. Using this as a definition, f_2 is a quasihomomorphism and extends f_1 . $\square$

Lemma 24. Let $f: [-b, b] \rightarrow \text{Arc}_{-1}$ be a map such that $f(0) = 1$.

1. If f is a midpoint homomorphism then it is a quasihomomorphism.
2. If f is a quasihomomorphism with image in $\overline{\text{Arc}}_i$ then it is a midpoint homomorphism. Hence, it is uniquely determined by $f(b)$ (and $f(-b)$, but that is $f(b)^*$).

Proof. (1) We evaluate $f((\theta + \phi)/2)$ in two different ways, then squaring both sides gives the result.

$$\begin{aligned} f\left(\frac{\theta + \phi}{2}\right) &= f(m(\theta + \phi, 0)) = m(f(\theta + \phi), 1) = \sqrt{f(\theta + \phi)}, \\ f\left(\frac{\theta + \phi}{2}\right) &= f(m(\theta, \phi)) = m(f(\theta), f(\phi)) = \sqrt{f(\theta)}\sqrt{f(\phi)}. \end{aligned}$$

(2) First, for each θ , $f(\theta) = f(\theta/2)^2$, so $f(\theta/2)$ is a square root of $f(\theta)$ in $\overline{\text{Arc}}_i$. By Lemma 9 there is a unique such, namely $\sqrt{f(\theta)}$.

Now for each θ, ϕ we have

$$f(m(\theta, \phi)) = f((\theta + \phi)/2) = f(\theta/2)f(\phi/2) = \sqrt{f(\theta)}\sqrt{f(\phi)} = m(f(\theta)f(\phi)).$$

$\square$

6 The arcsin map

In this Section we discuss the integral formula for arcsin. Its main application is in Section 8, where we deal with calculus, but it is also useful in providing an inverse for sin.

Definition 25. If $-1 < y < 1$, then we define

$$\arcsin y = \int_0^y \frac{dt}{\sqrt{1-t^2}}.$$

(Presumably this can be extended to $y = \pm 1$, but we would first have to develop a theory of improper integrals.)

Proposition 26. Suppose z_1 , z_2 and $z_1 z_2$ are all in Arc_i (so their imaginary parts are all in $(-1, 1)$). Then

$$\arcsin \Im(z_1 z_2) = \arcsin \Im z_1 + \arcsin \Im z_2.$$

Proof. We must show that³

$$\int_{\Im z_1}^{\Im(z_1 z_2)} \frac{dt}{\sqrt{1-t^2}} = \int_0^{\Im z_2} \frac{dt}{\sqrt{1-t^2}}.$$

Consider the map $z \mapsto z_1 z$ on S^1 . Transforming to the imaginary axis, let us write $u = \Im z$, so $z = \sqrt{1-u^2} + iu$. Then the map transforms to $u \mapsto t = (\Re z_1)u + (\Im z_1)\sqrt{1-u^2}$. By [Vic23] we can differentiate this using the usual rules including the power rule, giving

$$\begin{aligned} \frac{dt}{du} &= \Re z_1 + \Im z_1 \left(\frac{1}{2} \right) \frac{-2u}{\sqrt{1-u^2}} = \frac{(\Re z_1)\sqrt{1-u^2} - (\Im z_1)u}{\sqrt{1-u^2}} \\ &= \frac{\Re(z_1 z)}{\sqrt{1-u^2}} = \frac{\sqrt{1-t^2}}{\sqrt{1-u^2}}. \end{aligned}$$

Hence

$$\begin{aligned} \int_{\Im z_1}^{\Im(z_1 z_2)} \frac{dt}{\sqrt{1-t^2}} &= \int_0^{\Im z_2} \frac{1}{\sqrt{1-t^2}} \frac{\sqrt{1-t^2}}{\sqrt{1-u^2}} du \\ &= \arcsin \Im z_2. \end{aligned}$$

□

Definition 27. The real number π is defined as $4 \arcsin(1/\sqrt{2})$.

Proposition 28. $e_{\frac{1}{8}}$ (Definition 21) is invertible, with inverse

$$e_{\frac{1}{8}}^{-1}(z) = \frac{\arcsin \Im z}{2\pi}.$$

³Essentially we are showing that $dt/\sqrt{1-t^2}$ on the imaginary axis gives a translation invariant measure on (part of) S^1 : in other words, we are on our way to defining a Haar measure.

Proof. The map $\arcsin$ is strictly monotone, and hence, by Lemma 4, is a homeomorphism to its image. Then, using the definition of π , we obtain homeomorphisms

$$\overline{\text{Arc}}_{\sqrt{i}} \xrightarrow[\cong]{\Im} \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right] \xrightarrow[\cong]{\arcsin} \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \xrightarrow[\cong]{\theta \mapsto \theta/2\pi} \left[-\frac{1}{8}, \frac{1}{8}\right].$$

By Proposition 26 the composite of the first two transforms products to sums, insofar as they exist in the domain, and we deduce that the inverse of that composite is a quasihomomorphism. $\square$

7 The exponential maps $E_a(\theta) = e^{2\pi i a \theta}$

In this section we describe the exponential maps $E_a: \mathbb{R} \rightarrow S^1$, $\theta \mapsto e^{2\pi i a \theta}$. One of them ($a = 0$) is the constant map 1. For the others, our main focus will be E_1 , which maps $[0, 1]$ once round the circle – its kernel is the integers – and for which $E_1(1/4) = i$. In effect, its argument is measured in angular units of revolutions. For a unit of a revolutions we define $E_a(\theta) = E_1(a\theta)$, so for θ in degrees or radians we use $a = 1/360$ or $a = 1/2\pi$. For $a < 0$ we go backwards.

Definition 29. Using Lemma 23, we can extend $e_{\frac{1}{8}}$ (Definition 21) uniquely to quasihomomorphisms as follows:

$$\begin{aligned} e_{\frac{1}{4}} &: \left[-\frac{1}{4}, \frac{1}{4}\right] \rightarrow \overline{\text{Arc}}_i, \\ e_{\frac{1}{2}} &: \left[-\frac{1}{2}, \frac{1}{2}\right] \rightarrow S^1, \\ e_1 &: [-1, 1] \rightarrow S^1, \\ e_2 &: [-2, 2] \rightarrow S^1. \end{aligned}$$

e_2 wraps $[-2, 2]$ four times round the circle, with $-2, -1, 0, 1$ and 2 all mapping to 1.

Theorem 30. *There is a unique homomorphism $E_1: \mathbb{R} \rightarrow S^1$ that extends $e_{\frac{1}{8}}$, whose kernel is $\mathbb{Z}$ and such that $\Im E_1(\frac{1}{4}) > 0$ (in fact, $E_1(\frac{1}{4}) = i$).*

Proof. If E_1 exists, then $E_1(1) = 1 \neq E_1(\frac{1}{2})$ and so $E_1(\frac{1}{2}) = -1$. It follows that $E_1(\frac{1}{4}) = i$.

Suppose $\theta \in \mathbb{R}$. If ε is a small, positive rational, then we can find a rational q such that $q - \varepsilon < \theta < q + \varepsilon$. Let $n = \lfloor q + \frac{1}{2} \rfloor \in \mathbb{Z}$, so $-\frac{1}{2} < q - n \leq \frac{1}{2}$ and

$$-\frac{1}{2} - \varepsilon < \theta - n < \frac{1}{2} + \varepsilon. \quad (2)$$

If E_1 is to extend $e_{\frac{1}{8}}$ then it must also extend e_2 . Hence, given such an n , we must define $E_1(\theta) = e_2(\theta - n)$.

Suppose we also have n' satisfying the conditions (2) for n . Then

$$-1 - 2\varepsilon < n - n' < 1 + 2\varepsilon,$$

so provided $\varepsilon < \frac{1}{2}$, we have that $n' - n \in \{-1, 0, 1\}$. Then $\theta - n'$, $n' - n$ and $\theta - n = (\theta - n') + (n' - n)$ are all in $[-1, 1]$, so $e_2(\theta - n) = e_2(\theta - n')e_2(n' - n) = e_2(\theta - n')$. This shows that $E_1(\theta)$ is well defined.

(We can be more precise. Fixing θ , let W be the set of n such that condition (2) holds. This is a set, because the condition is geometric, and the argument shows that $W \rightarrow 1$ is a surjective local homeomorphism and hence the coequalizer of its kernel pair, which is given by the projections $W \times W \rightarrow W$. Our proof of well-definedness shows that the map $n \mapsto e_2(\theta - n)$ composes equally with those projections and so factors via 1.)

To prove the homomorphism property, first, it is obvious that $E_1(0) = 1$. Now suppose $\theta, \phi: \mathbb{R}$, with integers n satisfying condition (2) for θ and likewise m for ϕ . Then

$$-1 - 2\varepsilon < (\theta + \phi) - (n + m) < 1 + 2\varepsilon,$$

along with, for some integer k ,

$$-\frac{1}{2} - \varepsilon < (\theta + \phi) - k < \frac{1}{2} + \varepsilon.$$

From these we deduce

$$-\frac{3}{2} - 3\varepsilon < (n + m) - k < \frac{3}{2} + 3\varepsilon,$$

and so, provided $\varepsilon < 1/6$, we have $n + m - k \in \{-1, 0, 1\}$ and

$$E_1(\theta + \phi) = e_2(\theta + \phi - k) = e_2(\theta + \phi - n - m) = e_2(\theta - n)e_2(\phi - m) = E_1(\theta)E_1(\phi).$$

To see that E_1 has the right kernel, clearly the kernel contains $\mathbb{Z}$. Now suppose $E_1(\theta) = e_2(\theta - n) = e_{\frac{1}{8}}((\theta - n)/8)^8 = 1$. Using part (2) of Theorem 20, we see that we seek an eighth root of 1 in $\text{Arc}_{\sqrt{i}}$, and there is only one – namely 1 itself. Now using invertibility of $e_{\frac{1}{8}}$ (Proposition 28) we deduce that $\theta = n$.

By definition, $E_1(1/4) = e_{\frac{1}{4}}(1/4) = i$, with positive imaginary part. $\square$

Definition 31. The homomorphism $E_a: \mathbb{R} \rightarrow S^1$ is defined by $E_a(\theta) = E_1(a\theta)$.

Remark 32. One might conjecture that the condition “that extends $e_{\frac{1}{8}}$ ” in Theorem 30 is unnecessary, that E_1 is the unique homomorphism with kernel $\mathbb{Z}$ such that $\Im E_1(1/4) > 0$.

Even to show that $E_1(1/8) = \sqrt{i}$ is not trivial. Once one has $E_1(1/4) = i$, $E_{1/8}$ must be $\pm\sqrt{i}$, and for each one there are two homomorphisms that immediately present themselves. We get E_a for $a = 1, -7, -3$ or 5 , where $E_1(1/8)$ for the first two is $\sqrt{i}$, and for the last two is $-\sqrt{i}$. The final three are ruled out by having the wrong kernel, but it is not clear how to generalize that argument.

8 Differentiation

To develop the calculus rules, we follow a line similar to that of [Vic23]. There, the differentiability of the exponentials (homomorphisms from $\mathbb{R}$ to $(0, \infty)$) follows from the fact that their inverses, logarithms, have derivatives that do not rely on the exponential theory: $\frac{d}{dx} \ln x = 1/x$. We can then use integrals and exploit the Fundamental Theorem of Calculus (FTC).

Here, where we deal with homomorphisms from $\mathbb{R}$ to S^1 , we can use a similar trick based on $\frac{d}{dx} \arcsin x = 1/\sqrt{1-x^2}$.

Of course, the natural angular measure here is *radians*, and this fits with using $E_{1/2\pi}$ as exponentiation map. Now Proposition 28 tells us that

$$E_{1/2\pi}(\arcsin \Im z) = E_1\left(e^{-1/\frac{1}{8}}(z)\right) = z.$$

In line with this we define –

Definition 33. The maps $\sin, \cos: \mathbb{R} \rightarrow [-1, 1]$ are defined by

$$E_{1/2\pi}(\theta) = \cos \theta + i \sin \theta.$$

Immediately from the fact that $E_{1/2\pi}$ is a homomorphism we can deduce the angle sum rules (1). In addition, from the definition of π , we derive the following:

$$\begin{aligned} \sin \frac{\pi}{2} &= 1, \quad \cos \frac{\pi}{2} = 0, \quad \sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}, \\ \cos \theta &= \sin \left(\theta + \frac{\pi}{2} \right). \end{aligned} \tag{3}$$

Theorem 34. *The maps $\sin$ and $\cos$ are differentiable, with $\sin' = \cos$ and $\cos' = -\sin$.*

Proof. We first prove these on the interval $(-\pi/2, \pi/2)$, where $\sin$ has inverse $\arcsin$. From the FTC, $\arcsin$ is differentiable with derivative

$$\frac{d}{dy} \arcsin y = \frac{1}{\sqrt{1-y^2}}.$$

By the chain rule, it follows that $\sin$ is also differentiable, with derivative

$$\frac{d}{d\theta} \sin \theta = \sqrt{1 - \sin^2 \theta} = \cos \theta.$$

Now we can calculate

$$\frac{d}{d\theta} \cos \theta = \frac{d}{d\theta} \sqrt{1 - \sin^2 \theta} = \left(\frac{1}{2} \right) \frac{-2 \sin \theta \cos \theta}{\sqrt{1 - \sin^2 \theta}} = -\sin \theta.$$

Now suppose θ is in the range $(\theta_0 - \pi/2, \theta_0 + \pi/2)$. We calculate

$$\begin{aligned}\frac{d}{d\theta} \sin \theta &= \frac{d}{d\theta} (\sin(\theta - \theta_0) \cos \theta_0 + \cos(\theta - \theta_0) \sin \theta_0) \\ &= \cos(\theta - \theta_0) \cos \theta_0 - \sin(\theta - \theta_0) \sin \theta_0 = \cos \theta \\ \frac{d}{d\theta} \cos \theta &= \frac{d}{d\theta} (\cos(\theta - \theta_0) \cos \theta_0 - \sin(\theta - \theta_0) \sin \theta_0) \\ &= -\sin(\theta - \theta_0) \cos \theta_0 - \cos(\theta - \theta_0) \sin \theta_0 = -\sin \theta\end{aligned}$$

□

For angular units other than radians, we need to measure $\sin$ and $\cos$ using the imaginary and real parts of E_a , where a is the angular unit expressed as a number of revolutions. For example, for radians $a = 1/2\pi$, and for degrees $a = 1/360$. Since $E_a(\theta) = E_1(a\theta) = E_{1/2\pi}((a/2\pi)\theta)$, we obtain

$$\frac{d}{d\theta} \Im E_a(\theta) = \frac{d}{d\theta} \sin\left(\frac{a}{2\pi}\theta\right) = \frac{a}{2\pi} \cos\left(\frac{a}{2\pi}\theta\right) = \frac{a}{2\pi} \Re E_a(\theta).$$

Similarly, $\frac{d}{d\theta} \Re E_a(\theta) = -\frac{a}{2\pi} \Im E_a(\theta)$.

It follows that, for a general homomorphism $E = E_a$, a can be found by differentiating:

$$a = 2\pi(\Im E)'(0).$$

Note that for $a = 0$, so E_a is the constant homomorphism to 1, we get the expected zero derivatives.

9 Conclusions

This approach to trigonometry, with its detailed treatment of the unfamiliar setting of midpoint algebras, was unexpected to me. In its favour, it does seem to capture more directly the everyday intuition of wrapping a string round a cylinder. It is also probing more directly the infinitesimal structure of the manifold S^1 : it is covered by open, cancellative, iterative midpoint algebras whose structure, agreeing on the overlaps, is intrinsic to the manifold, and does not depend on a metric. Perhaps this leads to new insights into manifold structure more generally, although the obvious midpoint structure on curved geometry such as a piece of a sphere does not satisfy the transposition condition in Definition 1. That is a flatness condition.

A more conventional approach in constructive mathematics would be to use power series, as in Bishop and Bridges [BB85]. They start off with a general purpose technology of convergence (of Cauchy sequences) and power series. Then $\exp$, $\sin$ and $\cos$ are specified as solutions to differential equations, which leads to particular power series and the problem of whether they converge. Maybe that could be made to work geometrically, although it has its oddities. Computationally, one is unlikely to use a power series to calculate the sine of a large number, for example $\sin(10^6)$, and the computation expressed in Theorem 30 seems more realistic.

References

- [BB85] E. Bishop and D. Bridges, *Constructive analysis*, Springer-Verlag, 1985.
- [ES01] M.H. Escardó and A.K. Simpson, *A universal characterization of the closed euclidean interval*, Logic in Computer Science, 2001. Proceedings. 16th Annual IEEE Symposium on, 2001, pp. 115–125.
- [Har38] G.H. Hardy, *A course of pure mathematics*, seventh ed., Cambridge University Press, 1938.
- [MV12] Maria Emilia Maietti and Steven Vickers, *An induction principle for consequence in arithmetic universes*, Journal of Pure and Applied Algebra **216** (2012), no. 8–9, 2049–2067.
- [Sam25] Giovanni Sambin, *Positive topology. A new practice in constructive mathematics*, Oxford Logic Guides, no. 57, Oxford Science Publications, 2025.
- [Vic95] Steven Vickers, *Locales are not pointless*, Theory and Formal Methods of Computing 1994 (London) (C.L. Hankin, I.C. Mackie, and R. Nagarajan, eds.), Imperial College Press, 1995, pp. 199–216.
- [Vic97] ———, *Constructive points of powerlocales*, Mathematical Proceedings of the Cambridge Philosophical Society **122** (1997), 207–222.
- [Vic04] ———, *The double powerlocale and exponentiation: A case study in geometric reasoning*, Theory and Applications of Categories **12** (2004), 372–422, Online at <http://www.tac.mta.ca/tac/index.html#vol12>.
- [Vic05] ———, *Localic completion of generalized metric spaces I*, Theory and Applications of Categories **14** (2005), 328–356.
- [Vic09a] ———, *The connected Vietoris powerlocale*, Topology and its Applications **156** (2009), no. 11, 1886–1910.
- [Vic09b] ———, *Localic completion of generalized metric spaces II: Powerlocales*, Journal of Logic and Analysis **1** (2009), no. 11, 1–48.
- [Vic17] ———, *The localic compact interval is an Escardó-Simpson interval object*, Mathematical Logic Quarterly **63** (2017), no. 6, 614–629, DOI 10.1002/malq.201500090.
- [Vic22] ———, *Generalized point-free spaces, pointwise*, <https://arxiv.org/abs/2206.01113>, 2022.
- [Vic23] ———, *The fundamental theorem of calculus point-free, with applications to exponentials and logarithms*, <https://arxiv.org/abs/2312.05228>, 2023.